

Mindscape 308 | Alison Gopnik on Children, AI, and Modes of Thinking

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hello everyone and welcome to the mindscape podcast I'm your host Sean Carroll so here's a question given a problem how do you find the solution to that problem this is one of those questions that sounds deep or profound or something like that but in fact you might worry it is a little bit too abstract how can we hope to understand how in perfect generality to find the answer to a problem if you haven't given me a bit more information about what kind of problem it is you're talking about but it's not so abstract that we can't make some progress you know in general how do we in fact go about solving problems sometimes hopefully indeed you'll find a problem that resembles another problem a problem that you've seen before so you can either use or maybe adapt the kind of solution that you already knew about other times you know nothing about the context of a problem so maybe you'll just try some things randomly sort of flail about just get some information these kinds of questions as abstract as they are turn out to be frighteningly relevant to things like building artificial intelligence right when you turn on the computer and there's no software on it yet the computer doesn't have any pre-existing strategies for solving problems you have to choose how to build them in so how should we go about figuring out the best way in different contexts to solve problems well one thing to do is to look at what actual human beings actually do the lesson of today's conversation with Allison Gopnik is that human beings use different ways to solve problems in different life stages there's a way that we do it when we're adults when we're flourishing in the prime of life and there's a different thing that we do when we're children or babies very roughly speaking babies are a little bit more creative a little bit more free flowing they just try a whole bunch of things their attention is hard to pin down you might have noticed that if you've ever dealt with babies and that's a feature not a bug the fact that babies have trouble focusing their attention on something is a reflection of the fact they're trying to learn about the World by interacting with it in many different ways whereas adults are optimized for

something different hopefully by the time you're an adult by the time you're in your 30s you've learned a lot about problem solving strategies and you're more about perfecting the methods that you already know rather than flailing around randomly and learning new ones not that you can't do it not that it's impossible but you're better at some techniques than others and maybe who knows we sort of brush upon the possibility in the conversation that later in life once you're past your Prime uh reproductive cycle for example you can go back to being less hidebound I mean maybe that would be ideal we all know people who don't quite live up to that but you serve different purposes I I guess the overarching lesson here is that there's a division of labor not only between different human beings working in groups but even between different life stages of single human beings and as we'll talk about we do in fact learn some lessons that might be very very relevant to AI and programming computers and thinking about how we should best approach understanding the world that's what we're here to do here at mindscapes so let's [Music] go Alison gnik welcome to the mindscape podcast happy to be here so one of the things I guess we could start with from your work that that I've derived is the idea that little children kids babies whatever um maybe shouldn't just be thought of as unformed adults right that they actually think in a different way is that an okay way to put it yeah that's exactly the right way to put it so I think a lot of people have always thought about the 35-year-old um psychologist or philosopher is sort of the height of all of intelligence and then we just build up to that amazing person as we get older and then we fall off off as we get older but of course that doesn't make very much sense from an evolutionary perspective and in fact I think what's come become clearer and clearer is that children are really fundamentally a different kind of intelligence than than typical adults are um and there's also some interesting questions about Elders having a different kind of intelligence as well so it's more like a kind of tradeoff between different kinds of intelligences than it is having one magic thing which is called intelligence that we have more of or less of and that we have less of to begin with and more of uh more of later on and an idea that I've been working on a lot is an idea that actually comes from computer science which is the idea of a trade-off between different kinds of intelligences um and in particular a trade-off between what's called exploit intelligence and explore intelligence and what I've argued is that childhood is really about this kind of exploration intelligence which is not just different from but even intention with uh exploit intelligence and now I have to say when I first started making these sets of arguments I was as you could tell from that last comment I was a little snarky about those you know 35y old philosophers and psychologists um U sitting and thinking that they were the the Apex of

intelligence but since my own children have gotten older and I have more grandchildren I now I sort of reverse that I my main feeling now is oh my God those poor 3 5y olds the the children and the grand moms are getting to do all the really fun human stuff um what of my other slogans is that we're basically human up till puberty and after menopause and in between we're sort of glorified primates we're doing the things that all the primates do we're finding our way in the dominance hierarchy and we're mating and we're getting resources and all that stuff and it's only when we're little and we're old that we get to do things like theory of mind and discovery about the world and causal inference and cultural transmission and uh large scale storytelling all the things that really make us human so now I feel like okay we Grand moms and kids to just keep quiet the fact that we're having all the fun and then the 35 year olds are having to do all the work and so this is going to you know be a preview for what's to come in the in the conversation but that's basically because we're in that exploratory phase early on and and in what is nominally thought of as the prime of our adult lives we're more focused right we have tasks and we're very good at doing those tasks but we're less good at being flexible and creative about things so so there's this idea uh that comes from um computer science about trying to Sol what happens when you're trying to solve what's called a high-dimensional task that means that a problem that has lots of different possible solutions that vary along a lot of different dimensions and one thing you can do is you can just kind of tweak what you're already doing a little bit and see if that makes things a bit better and that's a very efficient effective way of trying to solve the problem so just change where you are a little bit see if that makes it better if it does then then try something else um and that's the essential exploit strategy that's the grown-up strategy here's the thing that I need to do I'm going to focus on trying to do it but of course the problem with that is that there might be another solution that's much further out in the space that's as it were you know in the far reaches of the box at least if not actually out of the box um and if you just keep making these little tweaks you're never going to get there but what you so what you could do is you could Bounce Around try lots of things sort of independent of whether they're you think they're going to be useful or not just just try things just see how the world Works um and in uh computer science they talk about this is the difference between a low temperature search think about it your your audience will will get this analogy right um think about it as if it was a molecule of air that's just either going very slowly or bouncing randomly around in the space and the interesting uh uh solution that comes out of computer science is that the best strategy so these two things trade off against each other right like you can't do both at the same time so which should you do and it

turns out the best strategy is start out with a big bouncy random wild search even though of course that's going to take a lot longer and you're going to spend a lot of time thinking about things that actually aren't going to help you and then gradually cool off to uh the more focused search and people in uh in computer science talk about this as simulated annealing it's like what happens when you heat up again your audience should know this what happens when you heat up metal and cool it to make it more robust um and what I think is that childhood is essentially Evolution's way of doing simulated annealing so if you think about those two descriptions there's one way that's focused and oriented towards outputs and then there's this noisy bouncy random kind of way behave you could sort of pretty immediately think which one fits your four-year-old better yeah and the thought is that that even though that might look from some perspectives as if it's a deficit because the kids are saying weird crazy things they're often strange pretend land they they do things that don't seem to make sense at first if you're in explore mode then that's really the thing that you want to do more than anything else so of of course in some ways what this means and we've actually shown this empirically is that children are more creative than grown-ups at least sometimes so for instance we've done a bunch of studies where we give children and grown-ups a problem that either has a pretty obvious there's a there's an obvious solution you could try or a much less obvious hypothesis one that it's to start out with is less likely and what we've discovered is that when the solution is the more likely the more obvious one as you might expect the grown ups are better at getting to the solution but when it's the unlikely hypothesis four-year-olds are actually better than say college undergraduates at getting to the right uh at getting to the right solution so in some ways the four-year-olds are are more creative than adults but the trouble is if you look at what we think of when we think about creativity and adults it's really got two pieces so it's not really easy to compare so if you look at what uh a creative solution for an adult the first part is generate lots of possibilities and then the second part is pick out the one that's good right pay attention to the ones that are good and not the ones that aren't and that first part generate lots of solutions think of lots of possibilities that's where the kids really Excel but for adults you also have to have the part about you also have to have the part about choosing the right solution in the end and that's the part where those executive uh executive capacities seem to be more useful so um the Great philosopher John lo I Sean I know you have the chair of natural philosophy who was a great natural philosopher um said that there were two faculties for human intelligence one was the faculty of wit and one was the faculty of judgment so the faculty of wit was generate lots of new ideas and the faculty of judgment was pick out which

ones were actually good ones and kids seem to have a tremendous amount of the faculty of wit not quite so much of the faculty of judgment fair enough but okay if we're giving Evolution credit for coming up with this you know nice division of uh temporal labor over our lifetimes let's let's face up to the fact that kids are pretty helpless like human babies they take a long time to be able to fend for themselves could you talk a little bit about how that fits in with what we know about other species and whether or not it's like an accident or actually optimizing something yeah it's interesting because um if you look across an incredibly wide range of species even including insects and there's some arguments that this is true even for plants but certainly true for all the primates and for uh and for mammals in general you see this really striking correlation between how smart perhaps from a human perspective but how how good an animal is at learning how large its brain is how much it uses its brain um and how long a childhood it has um so you see it the first the first context in which people notice this was with birds so if you look at birds like Corvids or crows very very smart birds um they spend as long as a year or two being fledglings needing to be taken care of whereas if you think about like the domestic chicken chickens are mature within a couple of weeks and although this gets me in trouble with Chicken Lovers and everywhere they're not very bright there or a better way of putting it is they're extremely good at doing the things that they do well like pecking for grain and they're very good at doing those from the time they're born they're not so good at learning and if you think about it from this sort of explore exploit perspective that kind of makes sense so if you're going to be a creature that lives in a changing unpredictable environment that has to do lots of new things um you're going to need a period to be able to do that exploration before you can exploit and that means that in that period you're going to be really bad at doing things you're going to be helpless and you're going to need other you're going to need the other adult members of the species to look after you make sure you stay alive make sure you have enough calories to to um uh you have enough calories to fuel All of This ferocious learning that's going on um and that seems to be sort of The evolutionary strategy I the calories are kind of a nice uh a nice part of this because it turns out that if you look at how many what percentage of your calories your brain uses up even as an adult you know 20% of your calories are going to your brain which means that your brain is you know an expensive Computing Gadget but when you were four 60% in fact almost 70% of your calories were going to your brain so basically you know your average 4-year-old is like like a you know something out of Doctor Who or Science Fiction it's basically this giant hungry brain that's going around the world hypnotizing us into feeding it both a bunch of

peanut butter sandwiches so its brain can work and a bunch of data so its brain can work so that's interesting because you're pointing to an explanation that is not purely physical or even mostly physical it's not about brain size or anything like that but that uh if we allow ourselves to poetically um attribute forces of natural design to intelligent design rather to Evolution um the this sort of planned difficulty in taking care of oneself is actually serving the purpose of letting the kids explore be creative and that's supposed to pay off later in life that's right so the the thought is and you know of course as always in biology and evolution it's complicated and it's connected to other kinds of aspects of animals and there are smart animals who don't have a long childhood like uh um cephalopods octopuses for example um uh different kinds of animals seem to solve this explore exploit problem in different ways sometimes insects and uh for instance seem to do it by kind of division of labor among some ants are some ants and bees are explorers are scouts and some of them are workers who are actually doing things so there's different ways of solving it but it does seem like this is a strikingly general and common solution among a very very wide range of of species and that suggests that it's got this adaptive it's got this adaptive value is this division of labor thing um a general Fe teacher I I think I remember it was Michael M Krishna who I had on the podcast and who mentioned this experiment which I guess is famous among psychologists where H uh human children and chimpanzees were asked to poke sticks into a box and get a reward and then when it's revealed that one of the sticks isn't doing anything uh the the children didn't learn oh sorry the yeah the children didn't learn and the chimpanzees did because the children were trusting the grown-ups right yeah and of course that's another dimension of having this long uh having this long childhood and having this particular kind of intelligence which is that we are also social and cultural Learners so we pass on information from one generation to another in a way that no other other animals do it more simply but they we do more of it than any other animal does and again a lot of that social transmission is taking place in the context of you know caregivers people who are who are moms and dads who are helping the kids but it's interesting because that is a famous experiment but it's worth pointing out that um again there's this trade-off so if you just imitate what the other people around you are doing then there's kind of no point because you're not going to make any progress um so somebody along the line has to actually innovate as well as imitating and there's a bunch of work we've done some of it and other people have as well that show that children are when they're imitating they're kind of balancing well what do I think about how this actually works and what do I think about the person who's actually uh demonstrating it to me so for example there's a lovely experiment

where if you have the experimenter says I'm going to show you how this works the children are much more likely to do the kind of overimitation that you've described so they'll just do what the experimenter did but if the experimenter says gee look at this I don't know how this works do you know how this works then the children are much more likely to explore including exploring in these kind of unlikely ways and vice versa we did some experiments that showed that if you actually show get let children discover something about a machine for instance a lot of our experiments depend on this wonderfully simple inexpensive machine the blinket detector it's a little box that lights up and plays music when you put some things on it and not others and it costs like I don't know \$29.99 or something it's a nice nice nice from the grant perspective uh and it's it's been it's it's been extremely productive so if you do something like present the child with a little box let them play with it and they see that say red blocks make it go and then you have an adult who says oh blue blocks make the detector go the children it's interesting they'll just come exactly split the difference between what they see themselves and what the adult says so they won't just rely on what they do see themselves they won't just rely on the adult but they they'll produce one um they'll produce one solution or another kind of in proportion to the probability of those two of those two solutions so I think the fact that we're social beings but that in the course of cultural and social evolution Michael might have talked about this we have to balance imitation and innovation and it's quite complicated to do that that I think that also plays a role in the special intelligence of children well it makes me think very much of being an adviser to students right uh especially graduate students where you want to say look please do be creative and come up with new ideas but please also read some of the literature and there's a balance there you don't want to just slavishly follow what other people have already done yeah I mean I think that the one of the interesting things that we're working on now um is thinking about caregiving thinking about what happens as you take care of someone else and that's been very that the and particularly think about the intelligence of caregiving thinking about how hard it is to figure out what you should do as a teacher or a therapist or a parent that will allow the person you're caring for to become autonomous right you don't want them to just imitate what you do you don't want them to just do the same thing but you also don't want them to get into trouble you don't want them to do stupid or dangerous things and how I think it's really interesting about how caregivers manage to walk that line no matter what kind of context uh they're in whether you're thinking about being a you know a grandmother of a 2-year-old or you're thinking about being a supervisor of post talks it's the same it's fundamentally the same

problem and it's having those caregivers that enables a lot of that cultural transmission to take place one of the interesting things that has come out that I've been writing and thinking about recently is that actually grandmothers seem to be playing a grandmothers and grandfather fathers I always talk about grandmothers egocentrically seem to be playing a particularly important role in that kind of cultural transmission so there's quite a bit of anthropological evidence that when you look at um you know who gives the songs who gives the stories who tells you the big aspects of the culture it's more likely to be older people than than parents so the parents are kind of trying to deal with the everyday keep the kids alive keep fed and it's actually the grandparents who are you know in my case telling telling um um reading naria books and singing old Broadway show tunes and and doing things that are things that are especially designed to serve this function of cultural transmission and of course just as we have this exceptionally long childhood we also have this long post-menopausal grandmotherhood elderhood uh which is not characteristic of other uh other animals except except interestingly for orcas so orcas are one of the few examples of a mammal that also continues to live for some substantial period of time after menopause and has grandmothers that live with a pod and what the grandmothers do is pass on information about what to eat and how to function so especially when things get get short the grandmothers are the ones who lead the Pod out to oh okay I remember you know 20 years ago there was Krill in this in this site and um and help to help the the grandchildren uh the children and grandchildren to survive and I think that's another really nice sense of a kind of intelligence that's different from what we think of as the standard grown-up intelligence that sounds almost too adorable to be true how sure are we that the uh that the killer whales really uh put so much responsibility in the grandmother's um hands what what I like about this is what the C what it shows is the really important thing to pass on is recipes right the thing that the grandmothers that's the thing that the grandmothers are designed by Evolution to to do and the the uh I I just again with the adviser thing just it's on my mind we can be egocentric a little bit that's okay um I do think a very very common mistake in dealing with students is that the older wise folks are sometimes too good at saying why a good why a new idea won't work right right and and I mean are you are you hinting that maybe I'll get better at that as I get older I'll be more open to the exploration well I think that's definitely one of the uh that's definitely one of the features of of Elders that seems to be important and you know again to use the analogy of the students I think part of the reason for that is that the 35y old caregiver also has their own agenda right so they want to make sure that they get the results that they need for the Grant and the more the

older person can sort of say okay I'm I've gotten what I need I can be um more open to the possibility that this younger generation is actually going to do well now of course there's big individual differences among grown-ups about how much How likely they are to to do that or not um but it's interesting that even if you there's a little bit of evidence that even if you look at adult scientists and you know one of my one of my other slogans is that it's not that children are little scientists it's that scientists are big children and I almost always get a round of applause you know if I if I say this at fery lab or or you know larence Berkeley lab everybody sort of agrees with that um uh and one of the the there's a little bit of empirical evidence that shows for instance that the labs that get The Nobel prizes are the ones where something unexpected happens so what happens when you do your you do your experiment and what comes out is like totally weird not what you expected at all and you could say okay something strange happened let's go back to the ground or you could say huh like why did that happen um and it turns out that the labs where when something unexpected happens they follow up and try and figure it out do better in the long run than the labs where you're doing the thing that you were supposed to do okay I will try to let go of graduate students as an example here and and think more about the actual human babies uh tell tell us more about how they get their picture of the world I know that one crucial step is when they figure out that other people have opinions or beliefs that are different than their own yeah so the the big idea about Child Development which now I and other developmentalists have been arguing for for the last I don't know 20 25 years is that you should think about um development as being like Theory formation in science and at first that idea seems sort of unlikely I mean little kids are they not like really smart scientists but when you look carefully at the way they understand the world it looks a lot like the kinds of theories and the changes in theories that adults uh that scientists have and what that suggests is that from a a kind of computational perspective that's a just a really good way of getting information about the world so what we've shown is that from the time they're very very little children are looking for causal relationships for example which is one of the things that real that's really important in a theory and we can show again with our little blicket detector these little machines that have where they have unexpected causal properties we can show that even you know toddlers are doing the right kinds of inferences about how those kinds of systems work so they seem to develop these causal models these kind of what are sometimes called intuitive theories um and then they get new data and they change the theories depending on the data and we can show that really systematically so we can we can for instance show them information data about how that little little um uh about

how that little uh Machine Works the little licket detector works and then we can see what kinds of inferences do they make into a remarkable degree they make the inferences that you should make if you're trying to if you were being a good scientist but of course you know figuring out how machines work is fine but the thing that's most important in all of our Lives arguably is figuring out how the people around us are working um so way back in the 80s I and others um did developed what's come to be called um theory of mind which is figuring out what is it in baby's Minds that enables them to understand your mind and what we found out which is sort of typical of what we found out is that babies both know more to begin with than we would have thought but they also learn more um so between say one and two for instance they seem to start learning that people could want different things I could want one thing you could want another between uh four and six they seem to start understand between three and six they start to understand that I could think something different than you do and those are really really deep important things to understand but more recently one of the things that I think has been very interesting is the development of work on children's intuitive sociology so they're also figuring things out like oh when people are allies they'll behave this way and ah the next level good yeah right when they're enemies then they'll behave this other way and if one uh creature is dominant is larger than the other then they can get their way and what we're just trying to do now again to go back to this caregiving point is figure out what do they actually think about caregiving and there's a little bit of evidence that even babies already are identifying okay this is what it looks like for someone to take care of someone else this is a good potential this is a good potential caregiver for me so there's this lovely kind of back and forth between that helplessness that I talked about and then the fact that if you are that kind of helpless creature learning about other people learning about love learning about what other people are like is going to be really important for your your survival and of course as adults that's still the most important thing that we learned about I want to get the in the audience's mind the idea of those blicket detector clearly it's playing a big role in your psychology experiments is this something you buy off the shelf or do you make them yeah it's funny we we started out uh when we started doing this 20 years ago um we actually made them so we had in those days in the psychology department you had a shop which we still T to have anymore but we had a shop and we went to the folks in the shop and said okay well look here's what we want we want this little it's just it's very simple it's a little box you put different things on top and sometimes it lights up and plays music and sometimes it doesn't and then we had various other kinds of variants of this so we had one with gears where you flick a switch and

then the gears go and then that make something else happen so they're kind of like uh rub little Rube Goldberg machines that we'd we'd uh we'd put together now at the time that we this is a kind of interesting um anecdotal observation at the time that we first did this we thought gee this would be so much easier to do if we just had screens in a computer and you could do it that way the kids didn't want to have anything to do with when it was a a screen they really needed to have the real thing but now what we're finding is that even like three and four-year olds um and I think the big difference is now screens are interactive so three and four olds have the idea and in fact again if you have a a baby you know really love the idea that they can act on uh uh they can touch something they can talk to it they can swipe it and then figure out what the outcome uh is supposed to be although it still looks like we need the real things for the for the younger kids but for the older kids the older kids now sort of have the idea oh okay a a a screen is something that has causal Powers as well as a a little machine does but still you have these little you still doing the little boxes I I just like the idea that it's a right of passage in the gnik lab for graduate students to build a little box that lights up under different circumstances well that's that's exactly right and we have a lab culture about where we get the doorbells that go in it and um one of the funny uh a funny story about this when we first started doing it um uh when we first started doing it we actually had the folks in the shop who were who were putting these things together and at one point one of the machines that we had broke and my brilliant graduate student and I went in and asked said look it's not working the you know the the gear is not making the other gear go and they said oh no that's not how it works we just programmed it so it does this specific thing when you so there's no there's actually no physics behind it at all it's just a it's like a you know it's like a Tesla car it's just computer masquerading as a a real physical artifact and we were I'm have to say we were a bit crushed about this also the way that it actually works most of the time in the lab is that there's a graduate student who's sitting and pressing a button underneath the table that's determining what so from the kids perspective too it's not they're they have the illusion of actually doing physics but what they're actually chewing is doing experiments It Feels Like The Wizard of Oz here I feel like I've been fooled about all this all this stuff going on okay so uh if you say things like you know a child under one-year-old doesn't have this theory of mind and a two-year-old does so do we know of certain benchmarks or phase transitions in the growth of a child where there you know cognitive view of the world expands a bit Yeah so as I say from the very beginning I mean literally from the time they're born babies are doing things like paying special attention to faces and interpreting them from very early on they're imitating

other people in a way that other primates don't seem to which suggests that there's something about understanding other people that's really important um by the time they're nine months old they're doing things like pointing to communicate and a 9-month-old will point and if you don't follow the point they'll get ansy and go you know dare dare dare um so that suggests that they know something about what the fact that other people are are looking at the same thing that they are um so sometimes what happens is that people say oh you don't get theory of mind until you're you're four or five but that's not right you just get different theories the same way you would if you thought about you know physics there's different kinds of theories about different parts of of the Mind um uh what but very characteristically you see these changes coming at you know round about a particular age so about nine months you see this big Revolution and you know some of it may just be maturation but I think the something that is increasingly important and that we haven't thought about enough is just how active Learners even these very young children are just how much what they're doing is not just observing statistics although of course they they do that but but also actively experimenting on the world and you know if you think about even a newborn they're looking at you they're smiling they're seeing what the effects of their actions are and by the time by the time you're talking about um uh toddler that's like their whole lives right they're just constantly constantly doing experiments except when they do it we call it getting into everything um when physicists do it we call it being a good experimental physicist um uh and I think it's really under so one of the things that we've been doing a lot recently is trying to think about how the children's learning compares to the learning of uh machines for example something like the large models that have had so much U uh play so what's the relationship between what AI is doing and what these very little kids are doing and I think one of the really big Dimensions that makes the kids different from their AI systems is that they are going out and actively trying to get information about the world and then changing what they think based on the information they get as opposed to the large models for example that are basically just it's it's interesting to think about this in terms of cultural transmission the large models I've argued really what they do is just pick out patterns in all the things that other people have already have already figured out so they're just doing the imitation part of of cultural transmission um but the kids are actually going out there and finding out things that are new and experimenting and seeing new outcomes and we think that may be really The crucial thing that lets them learn as much as they do as quickly as they do and with as few calories as they do compared to the compared to the AI systems so it is a little bit like the distinction in causal reasoning between simply

finding correlations versus being interventionist about things right going outside the data set and saying if I did this what would happen next yeah that's exactly right and uh um and we've for 20 years we've been collaborating with actually philosophers of science and computer scientists who've been trying to figure out how do we do causal inference in science and one of the really you know basic ideas is that you can't or it's very difficult to do it just by looking at patterns of Statistics or correlation that you want to make causal information as well what makes a causal inference different from just a correlation this is an old philosophical problem back to Hume and I think the idea that's become most prominent in philosophy of science and seems intuitively and in science itself is well okay if you want to really make the causal inference you have to do an experiment you have to actually intervene do something in the world see what the outcome is that's what makes something uh causal and I think we've been doing some work on this but I think if you look at uh even quite young babies um you know if you think about something like a busy box a busy box is a toy that you use for little babies they're called busy boxes a busy box is a thing that has lots of causal possibilities where you can do lots of experiments it's cheaper than your you know standard cyclotron but it has the same kind of character for babies it's a way of um it's a way of actually being able to do the kinds of experiments that you need to figure out how the world works and of course as I point out the since um theory of mind and intuitive psychology is so important they have a great experimental subject there all the time which is the caregiver so if you think about them as being little scientist little psychologists we're the labrats so a lot of what babies do the Terrible Twos for example is um doing something and then looking to see how we uh react to it you know we had uh Judea Pearl on the podcast and at one point yeah and at one point he said literally like haven't you ever seen a baby all a baby is doing is like touching things and making a causal map of the world and I understood what he was saying and agreed with it but I guess I didn't realize it was quite as literal as uh apparently it is not absolutely so what we've done is taken some of the formalisms that Pearl developed like HEBEL and then show how the children are uh how the ways that children are solving these causal problems are just like uh what you'd expect from causal graphical models so essentially what the children I mean I think we can say that we've demonstrated this what the children are doing is constructing causal uh graphical models from data and then using them to determine their interventions now the thing is that both for Judea and for the whole world of causal inference and for us it's still a kind of Unsolved problem about and for science isn't right how do you decide which experiment is really the right experiment to do sure so what

something like um um Juda uh uh framework lets you do is say okay if you get this result here's the causal graph you should have but how do you decide what you should test in the first place um and I think one interesting thing that's coming out is that a lot of the um the sort of old shth about how you do experiments you have to keep everything constant you can only vary one thing at a time that may that's not true for babies that it may not be the best way of doing science either it and and I think this is like a wonderful um open uh a wonderful open question about how do you decide what kind of uh experiments to do how do you decide how to explore and we've started doing this with kids in um kids and AI agents what we've done is put them in the same kind of General environment um uh Minecraft is one of the ones that we use if you know M I imagine your audience will know Minecraft although there's a bit of a funny story about this which is that um I originally started doing this work about how you explore the Minecraft environment because my grandson was mad about Minecraft and I figured well at least this will make me cool right if I can tell them the Grandma's doing stuff about Minecraft um then I gave a big talk about this I got something called The Rumble heart prod it's a big deal and my grandchildren were in the audience and I said well I did this because I wanted to impress Auggie and he came up to me very sweetly after the talk and said Grandma this is a little that was a little embarrassing because like nobody cool plays Minecraft anymore we all play we all play fortnite um so like it isn't really cool to be involved in in Minecraft if you're a 13-year-old so I I wasn't keeping up with the changes but in any case Minecraft is a great game it continues to be a great game for for grown-ups and for 11y olds and um and it's uh uh what we've done is put kids in that environment and then put AI agents in that environment and just said figure out what's going on explore and even relatively young kids will explore that environment in a rational way they're not just randomly trying things they're doing things in a way that lets them figure out how the environment uh how the environment works and they're much much much better at it than AI agents are interesting well the thing you said about traditional scientific methodology asking us to keep all but one thing constant and changing that one thing and seeing what the effective I mean that makes s that sounds sensible to me uh but is it just because it's simpler to keep track of what's going on but maybe it's ultimately less effective than kind of poking things in concert yeah so that's a really interesting technical formal question about when you know that it is true that if you did that that's a simple normative rule about what to do but sometimes actually doing trying to vary different things at once can be more informative than than because as you can imagine it's really hard to keep everything else constant and just change one thing um and that that I think

thinking about what the kids are doing can be very because the kids some the kids are are often trying lots of different things in different in different ways if you think about you know the 2-year-old with the busy box um and yet they seem to be such good Learners so it's a really interesting question I think for AI for science for developmental psychology about how that's possible and how literally true is it to describe these very young children as good basian yeah I think it is literally true um uh We've what we've done is take various kinds of ideas from Bayesian inference and what we can do is give kids information for instance about the Baseline probability of a hypothesis and then we can give them information that will let them update that hypothesis and they'll do the right kind of updating in the right kind of way there another kind of discovery that was in my lab but also even more in Labs like uh Richard Asand and Jenny Saffron's back in the 90s was that children are quite good at doing statistics which you might be really surprised at because after all grown-ups are notoriously bad at doing probabilities but if you do something like um show the children that one blicket works um eight out of 10 times and the other one works four out of 10 times um they'll pick the one that has a higher probability of activate and then this is with 18month olds and then you say okay which one will you use here make the machine go they'll pick the one that has the higher statistical probability in fact they'll do it even if it's like $2/3$ versus $8/10$ so they seem to be able to kind of do the math implicitly obviously and unconsciously to figure out what how probabilities work so you know there there's and but again this has led to this puzzle which is they really seem to be doing something like basian inference and yet we know that basian inference ideally is sort of impossible computationally so if you have a if you have a big set of hypotheses then it's going to take you you know if you the the basian idea is you take a hypothesis you generate a pattern of evidence you check it against the data that you've got you figure out which hypothesis would have been most likely to generate that pattern of data um and the problem is if you have a big space of hypotheses that's just going to take forever because you're going to have to try each one of the hypotheses separately it's called the search problem and nobody's really quite solved that problem either in statistic or in AI or in childhood and the children do seem to be able to solve that problem because they end up you know finding out about the world and it's a really interesting question about how they solve it and my guess at the moment is that something about this active inference might be part of the solution that it isn't just that you're sort of sitting back having data float over you and then trying to update your hypothesis you're actually out there being an experimenter and figuring things out um uh which should make at least the you I know that when in physics there's

always a bit of a tension between the theoreticians and the experimentalists so um I think from the developmental perspective we're we're team experimentalists that's the the fair enough that's that's that looks like that's really the secret and how much do we know about the Neuroscience of this kind of thing I mean are these ideas that babies sort of get better at different kinds of reasoning as they grow older mirrored in ways that the brain is wiring itself yeah it's quite neat because one of the oldest uh developmental Neuroscience findings which is consistent with what we've we've seen to this day is that um early in development what you see is lots of new synapses lots of new connections being formed and then there's a point a kind of Tipping Point where the ones connections that have been formed formed are strengthened they're myelinated they become more effective the ones that haven't been formed are pruned so the weaker ones just kind of disappear so you have this early brain that is very very flexible very good at changing in the light of new experience very plastic as neuroscientists say and then but not very effective right not very good at actually doing anything and then you have this later brain that's very good at doing things but much less good at changing much less with much less plasticity and again this is like another version of this explore exploit trade-off and empirically if you look at brain development you can see that depending on the domain that you're in you see this pattern of very early um early proliferation and then later pruning so if you look at the visual system for instance you see this Tipping Point at around 18 months so the visual system is kind of getting set in that early period and then it kind of settles in which is why uh if you have vision problems it's really important to correct them in a baby it's really important to correct them early if you look at the language areas it's like five or six where you start seeing this you start seeing this transition so it's when you've developed a language that then it becomes harder for you to learn another um uh learn a new language um and if you look at what sometimes called the executive function the prefrontal part of the brain that's the latest one and that's not completely getting set until adolescence for example um okay so you see this but you do see this General pattern of start out with lots and lots of connections and then at some point start strengthening and pruning and I I just hate to bring it back to Academia once again but this this comports with the idea that you know like the energetic young idea creators are the young faculty in postdocs right not the uh older senior faculty you've been successful for many decades yeah I mean I think it's an obvious question that lots of people have asked is well are there things that grown-ups could do that would make them as creative as the uh as the children and the first thing to say is even if you're thinking about a lab if you want to get grant money it's really

good to have people who are really really practiced and and know how that system works and are really good at and even if you forget about grant money just you know actually doing the experiments if you're a physicist for example is a giant you know Enterprise it's really takes a lot of focus and executive energy to be able to get the experiments to run um so I don't think you'd want you know like you don't want your your department chair to be this wild wildly creative person who's thinking up strange ideas all the time right you want them to be to be focused nevertheless I think it's interesting that and and relevant to this explor expl tradeoff that um when adults want to adults often will kind of have social institutions that allow them to switch back and forth between this um explore and exploit so sabbatical I think are a great example in the academic context right so we don't sort of expect scientists to be able to just do this on their own what we think is going off to a retreat going off to a sabbatical going off to a workshop those are all ways that you can kind of pull yourself out of the context that you're already in and and allow yourself to to do greater um exploration so you know it's a pain that we all spend so much time traveling as scientists but I do think there's a bit of an argument that being in a different place being in a different context often doing interdisciplinary work where you're you're you're forced to think about things in really different ways because now you're looking at biologists instead of physicists all those seem to be ways that grown-ups set themselves up to be more creative and does it you've mentioned already several times the connection between um these ideas and AI large language models Etc how operationalizable are these insights can we imagine building better AI models using this uh the lessons we've learned from development well that's kind of what we've been doing so um I'm part of the Berkeley AI research group and um and what we're trying to do is take some of these ideas IDE about Active Learning ideas about social learning ideas about causal model build building and actually Implement them in in the in in AI systems and you know to a certain extent what's happened is that because the large models have been so effective in lots of ways um they've there's a tendency to sort of say okay well they worked before we'll just keep make we'll just you know pour more uh compute into them and they'll get bigger and they'll use up more energy and then we'll they'll get better and better and we'll have this mythical thing called AGI and one of the things when I talk to AI groups which I do a lot now I have a um I have a um uh a slide where I say there's no such thing as general intelligence artificial or natural which always leads to Sharp and take of breath from uh from the audience because it isn't the the the theory the the ideology the model and oh okay there's this one this gets back to our very first conversation the model that there's this one thing called intelligence some people

have more of it some people have less of it if you have more and more of it then you're going to be more and more effective right that's not the model that comes out of cognitive science at all what comes out of cognitive science is that we have these trade-offs between different kinds of cognitive capacities so if we could design sorry so llms don't rather strikingly have any of these Capac they're not going out in the world and doing experiments um they're not creating um abstract caal models that then they can use to generalize and you can see the difference between the amount of data that the children have and how good they are at generalizing it to new situations and and the large models the large models need enormous amounts of data and they're not very good at generalizing especially to what's called out of distribution cases cases that weren't there in their training the kids are ve with much less data are very good at doing that so the question is what what are the kids doing and could that make more effective Ai and I think the thing to say at the moment is we're very very if you start thinking about the comparison to kids we're very very far away so um they're they're the kids are you know orders and Orders of magnitude better than any of the current AI uh systems that we have but I think that's the direction that we might want to go in well it's a difficult conversation to have sometimes because the AI um the people who worry that we're close to Super intelligence any moment now will say look the you know we can beat the best human beings at playing chess or playing go or whatever and and you try to make the point as as you just did that there are other aspects of intelligence that they're not good at but it's they seem a little bit fuzzier right it's it's quite it's a little bit harder to put a benchmark on it is that something that we're trying to make progress on well I mean one thing is that uh for example if you look at robotics um the you know notoriously the the uh the AIS are better at playing chess but they're terrible at actually picking up the pieces if you if you made part of the game be that you had to pick up find the pieces in a you know that have spilled on the floor and put them back in exactly the right places they're really quite bad at doing that especially if you had say a new chess set that was uh you know like a an Allison Wonderland chess set or something where where it wasn't you couldn't you couldn't just identify them from your previous training um and this is the famous maravich's Paradox this goes back to beginnings of AI that uh things that look really hard for hum chess and go turn out to be relatively easy for AI and things that look really easy like perception and action and movement and the kinds of things that kids do when they figure out how the little blicket detector Works turn out to be really hard for um for AI I I think the whole conversation about intelligence in AI is fundamentally misguided because really the reason why the systems are as effective as they are is because they're taking

advantage of you know hundreds of thousands of humans who've done things like put text on the web and uh um and in some ways like with reinforcement learning from Human feedback have trained up the systems um my my latest metaphor about this is do you know the story of Stone Soup the old children's story of Stone Soup yeah this is a wonderful children's story this is another advantage of being a developmental psychologist is um you have wisdom of the stories that grandmothers have told in the past so this turns out to be a very very widespread bit of folklore um and here's the story there's these visitors who come to um a village and they say we want some food in the villagers say oh sorry we don't have any we can't share any with you and the visitors say that's okay we're going to make stone soup we have magic Stones so they get a big cauldron they put this couple of stones in The Cauldron it starts boiling they say see we're just going to make stone soup it's so good you know it would be even better if we had an onion and a carrot that we could put in it um but I guess if we don't have it we don't have it so one of the Villager says I I think I have an onion and a carrot somewhere they go and put it in they say oh that's great like oh see how well this soup is working um when we made it for the king we put a chicken in and that was really great so could you go and and of course you can guess what happens the the villagers all contribute their bit of of uh their bit of food and then at the end the villagers say this is amazing this is Magic we got all this soup just from a stone and I think if you think about a version of that where you said there were the computer people um The Tech Guys who went to the universe of computer users and said oh we have AGI we can do it just with you know three algorithms gradient descent and Transformers and we'll have AGI any minute now um but you know we really need a lot of data to make this AGI work and the users said oh okay we have all of our text and pictures that we put on the internet all of our books all of our newspapers um we could give you that to make the intelligence and then the the villagers the The Tech Guys say uh oh that's really great but like if we could do reinforcement the our AGI is still saying stupid things so if we could get reinforcement learning from Human feedback and and get people to give us feedback about whether our AGI is doing well or not that would make it even better um and the people said oh okay like there's whole villages in Kenya who could do this and finally the tech gu said oh that's really we're getting really good intelligence but if we could do prompt engineering so we could figure out what exactly what prompt to use that would make it even more intelligent could you do that and the user say oh yeah we could think a lot about that and then of course in the end what happens is that the the tech gu say see we told you we made AI just from a couple of algorithms um and it's ignoring the fact that the reason why it works is

not because of the algorithms but because of the data and that data comes from a bunch of humans who are doing the kind of exploratory creative intelligence that four-year-olds do I do presume that there are other angles on building AI systems that that are more oriented towards making causal models of the world even intervening putting them in robots I really I really don't know yeah that's one of the things that people have thought about but as I say it's hard so you know trying to do the basian inference is hard because the search problem is hard um there's what's called reinforcement learning and reinforcement learning old idea in Psychology is the and actually take actions and then see if the actions make you better off or not so you know the mouse who runs through the Maze and um um uh gets cheese sometimes but gets a shock other times right um and learns to go towards the cheese and away from the shock so that's a technique that for example in the um go uh solution reinforcement learning played a really important uh role in in trying to solve that problem so the chess uh the the chess agent is playing against itself over many many trials to try and figure out what the what the best reinforcement learning is but the problem with reinforcement learning is that it it's too narrow like if you just end up on you could since all you're thinking about is am I better off or not you're not going to be able to do the kind of exploration that you need to really figure out how the world Works um and so the idea that I think is most promising and we've been working on is an idea that should be very familiar from science which is an idea that you could do learning but instead of instead of trying to calculate whether you've got you know higher usefulness or utility or not or whether you got more cheese what you're calculating is did I get more information did I find out more about the world or did I find out more about the system when I when I did this action versus this other action and the result is that you are going to do again to go back to the point we made before you're going to do the things that will tell you something about how the world works even if they don't make you any better off in the in the short run um and there are ideas in uh an idea I really like and that we've been working on is the idea of a kind of intrinsic reward like the kind of reward you get as a scientist when you when you just are doing it because it's so cool when things work out the way that you want so I think the solution to the problem means having a kind of reinforcement learning uh agent but one where the reward isn't cheese or utility or winning the game it's finding out something new figuring out more about how the world works and there's been a lot of interesting attempts both in development and in AI to figure out how you could make that happen what kind of signal would tell you that as a result of this action you'd learn more and one that we're working on now that I think is really interesting is something called empowerment and the idea behind

empowerment is that you get rewarded when you do something and it has a predictable effect on the world so what you do is try to do as many things as you can where varying the action that you take will vary what happens out there in the world and when you do that that's really exciting that's really cool that is something you're going to be likely to try again but you also want as many different kinds of actions as many different relationships like that as you can so if you find one after a while you'll get bored and you'll try and find uh you'll try and find another one and I think that's very closely related to the causal learning that I talked about before so if you think about what you mentioned this Sean that causation is all about intervention right so really what you're learning with this empowerment reward is I'm going to learn how to intervene I'm going to learn how doing things out there in the world ends up having ends up having um uh effects and we have a little bit of evidence that even tiny babies you know if you take literally a two or three-month-old and you put a ribbon between their foot and a mobile so that they can actually control what happens to the mobile they'll sit there and try all sorts of different patterns of kicking to try and see what the consequences are going to be for the mobile so they really are like little and they'll smile and they'll giggle and they'll go back to doing it it just seems to be an incredibly satisfying uh an incredibly satisfying Enterprise for them and of course they're doing the same thing with Mom when they're you know doing all sorts of funny faces and seeing if mom gives funny faces back again um so I think that might be a really really important part of the solution and again it speaks to science because going back to what we were saying before about how systematic do your experiments have to be um one of the things that I've noticed is that the powers that be always say well you don't want to just have a fish fishing Expedition right this grant is just a fishing Expedition but a lot of times what you actually end up doing is doing a fishing Expedition you try something and then uh kind of unexpectedly it has this particular systematic result and that's the that's the where the gold is rather than the causation that you already that you already know about and empowerment is kind of a way of saying yeah get keep that fishing expedition going well and there are better and worse ways to go fishing right exactly exactly you know we did have Carl Friston also on the podcast and he has these ideas of the free energy principle and the basian brain and and part of it part of the sort of aspect that gives people pause is he seems to be saying that our brains try to minimize being surprised and you might think that therefore you should just sit in a dark room and never do anything but if I understand correctly he was saying no actually we want to minimize the net surprise over our lives and therefore we better explore around and do weird things now so we can anticipate what's coming yeah

that's exactly like the exploration exploitation um trade-off that I was talking about so it's it's interesting if you think about something like uh empowerment or all of these kind of intrinsic rewards the failure cases are you just sit there and do the same thing over and over again and maybe those are human failure cases too um so you need to do is to be able to want to have a coherent story about what's going on around you you don't want to just be uh you don't want to just be paying attention to random stuff that's happening yeah um even though that might give you something that's new and surprising um but you also don't want to just do the same thing over and over again and and we've been looking at you know comparing for instance what do children think when you give them a machine that just randomly does different things like our our blicket detector but a kind of random blicket detector versus one where there's a systematic relationship between what you do and what comes out even if what comes out is surprising versus one that just does something obvious that you know about beforehand and they really seem to like to play with the machine that that is surprising but surprising in a systematic way and I think that's true about scientists too the world has some structure and we do take advantage of that and I think this is something that we haven't quite uh as philosophers physicists AI researchers whatever uh learned how to systematize perfectly yeah I mean one of the things that that always strikes me I mean the reason why I started doing this work in the first place back when I was a philosophy student completely a philosophy student is if you think about um you know going back to Plato and Aristotle one of the deep philosophical questions is this question of knowledge it's how we know that there's a world out there it has structure not only does have it has you know quirks and minds and all sorts of things that we can't immediately observe in it and yet all that reaches us from that world is a bunch of disturbances of air at our eardrums and photons at our eyes how do we ever at any point manage to reconstruct that world from that data and I think going back to Plato and Aristotle the two ways of trying to answer that question have been to say well okay it just looks as if we're understanding that world from data it's really there all along so it's really some kind of innate there's some kind of innate evolved structure that's responsible uh for this and that's been one that's Plato's approach that's that's one of the approaches the other approach going back to Aristotle which is the approach of the most recent version of AI like the llms is it just looks as if we're really understanding the structure all we're doing is pulling out correlations between those photons and those disturbances of a so all we're doing is just taking the data and pulling out correlations and we think that that's telling us something about structure but but we have no particular reason to think that and the great thing about development is that if you look at

actual babies and children actually learning they don't seem to fit either of those pictures they seem to be able to learn really and I think also if you look at science um we seem to be able to learn really radically new things about the world whether it's learning that I might like broccoli and you don't or whether it's learning about uh you know uh corks and and leptons um and and yet it doesn't look as if all we're doing is just pulling together the finding statistical correlations in the data it looks as if we're genuinely developing theories that go beyond just the correlations in the data and I I don't think we have a good it you know even though we've been doing this for a thousand years we still don't have a good formal model computational model good understanding of how that's possible and I think looking at the kids who are clearly doing it is is a really good route to answering that philosophical question and that's what I've spent my whole career do it you know sometimes looking at what actually happens in the world does help us understand it better Alison gnik thanks so much for being on the mindcap podcast well thanks so much for having me Shan a great conversation